

Tezpur University

Spring Semester Mid Term Examination, 2022

BT 201: Biology for Engineers

Marks 40

Time: 2 hr.

Q1. Answer the following two questions:

[5X2=10]

- a) Differentiate Gram-negative bacterial cell wall is more complex than gram-positive cell wall bacteria? Justify.
- b) Draw a well labelled diagram of gram positive and gram-negative bacterial cell wall?

Q2. Choose the correct options and write it.

[2X5=10]

- a) Which is the correct representation of the hierarchy of life forms:
- Atom > Molecules > Organs > Organelles > Cell > Organisms > Population
 - Atom > Molecules > Organelles > Cell > Organs > Organisms > Population
 - Atom > Molecules > Organelles > Tissue > Cells > Community > Population
 - Atom > Molecules > Cell > Organelles > Organs > Organisms > Population
- b) On the basis of ammonia excretion the classification status given to shark is:
- ☒ Aminotelic
 - ☒ Uricotelic
 - ☐ Ureotelic
 - ☒ Both (b) and (c)
- c) Classification of an organism should not be based on:
- Cellularity
 - Cell structure
 - ☒ Gross morphology
 - Phylogenetic relationship
- d) Parasitic and saprobiotic modes of nutrition are found in
- Fungi
 - Bacteria
 - ☒ Both (a) and (b)
 - Ticks
- e) The purpose of taxonomy is to:
- Explain the origin of life

- ii. Identification of unknown species
- iii. To search the history of evolution
- iv. To determine number of organisms.



Q3. Filled up the gapes and rewrite the complete sentence

[1X10=10]

- a) Prokaryotes are grouped in Kingdom..... Monera
- b) ...Sexual.....reproduction is more conducive to driving evolution.
- c) An organism to be considered as model organism should have following two characters.....reproduce easily.....and... cheap.....
- d) One of the major difference among eukaryotic and prokaryotic cells is in having...~~nucleus~~ cell wall
- e) Uricotelic organisms excrete...ammonia
- f) Holotrophic organisms break down food...inside.....their body.
- g) Mode of reproduction of unicellular organism is primarily...asexual.....
- h) During the classification based on cellularity, division of labor is observed in...multicellular organisms
- i) Organisms using inorganic substrates as a source of electron donors to drive energy acquisition are called.....
- j) Symbiotic organisms are classified under heterotrophic mode of nutrition. ☒

Q4. State whether following statements are true or false (write question no. and True or False only, do not write the complete sentence)

[1X10=10]

- a) Energy cannot be transferred and transformed, but it can be created or destroyed. The first law is also known as the principle of conservation of energy.
- b) Every energy transfer or transformation decreases the entropy of the universe.
- c) Energy can be associated with the relative motion of objects; this energy is called potential energy.
- d) Catabolic pathways consume energy to build complicated molecules from simpler ones; they are sometimes called biosynthetic pathways.
- e) The totality of an organism's chemical reactions is called anabolism.
- f) Protein can be converted to DNA.
- g) In the genetic code, there are 64 triplet codons and 24 amino acids.
- h) RNA contains adenine (A), cytosine (C) guanine (G), and thymine (T).
- i) The genotype of an organism is what we see with our eyes at the macroscopic levels.
- j) The top-down approach can study biological complexity from phenotype to genotype levels.

cell wall is Thick
multilayered

Cell wall thin
single layer